An Additional Scalar Equation of Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity: Gravitational Pressure, Characteristic Curvature Length, and Strong-Field Four-Acceleration

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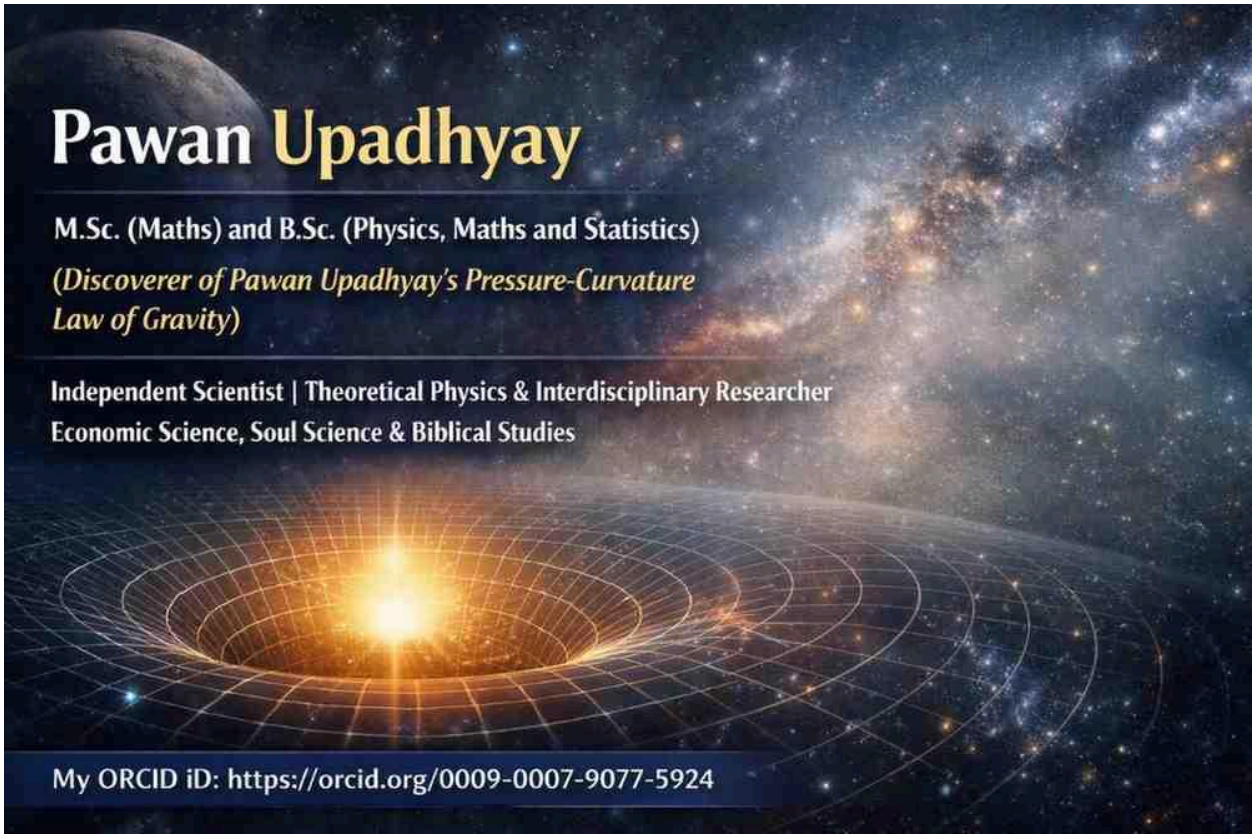
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Abstract

Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity proposes an interpretation of gravitation in which gravitational pressure, energy density, pressure gradients, and spacetime curvature play central roles. The existing PPC framework includes the relations $E_d = \rho c^2$ and $P_g = \omega E_d$.

This paper presents an additional scalar equation proposed within the PPC framework:

$P_g = P_0 + \rho a L$

Here, P_g is gravitational pressure, P_0 is a reference pressure, ρ is mass density, a is the relevant acceleration magnitude, and L is the characteristic length of curvature. The equation is presented as a distinct scalar relationship, not as a result derived from the tensor formulation or from the stress-energy conservation equation.

The paper considers the equation in weak-field and proposed strong-field applications. In the weak-field regime, a may represent the magnitude of Newtonian gravitational acceleration. In the strong-field regime, the author proposes the use of four-acceleration within a separate Energy Conservation Law formulation. The scalar pressure equation and the tensor formulation

are therefore treated as mathematically distinct components of the broader PPC research programme.

The paper defines the variables, examines dimensional consistency, explores the relationship between the additional equation and the existing PPC pressure relation, and identifies the mathematical and physical questions that must be investigated before the proposal can be regarded as an established law of gravitation.

Keywords: Pressure–Curvature Law of Gravity, PPC, gravitational pressure, characteristic curvature length, scalar equation, energy density, four-acceleration, energy conservation, weak-field gravity, strong-field gravity.

1. Introduction

Gravity has traditionally been studied through Newtonian mechanics and Einstein's General Theory of Relativity. Newtonian mechanics describes gravitational acceleration in terms of a gravitational field, while General Relativity describes gravitation through spacetime geometry and the relationship between geometry and energy-momentum.

Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity proposes an alternative physical interpretation in which gravitational pressure and its distribution are central to understanding the relationship between matter, energy, and spacetime curvature.

The PPC framework includes the following relations:

$$E_d = \rho c^2$$

$$P_g = \omega E_d$$

Combining these relations gives:

$$P_g = \omega \rho c^2$$

In these expressions, E_d represents energy density, ρ represents mass density, c represents the speed of light, P_g represents gravitational pressure, and ω represents the relevant proportionality or equation-of-state parameter within the proposed model.

The additional equation investigated in this paper is:

$$P_g = P_0 + \rho a L$$

This equation introduces a reference-pressure term and a characteristic curvature length. Its purpose is to express the proposed gravitational pressure in terms of mass density, acceleration, and a characteristic length associated with curvature.

An important clarification is necessary: this additional scalar equation is distinct from the tensor formulation of PPC. It is not claimed here to have been derived from the stress-energy tensor or from the covariant conservation equation. Its mathematical consistency and its relationship to the other parts of PPC must be investigated separately.

2. Objectives

The objectives of this paper are:

1. To present the additional scalar equation $P_g = P_0 + \rho aL$.
2. To define the physical quantities appearing in the equation.
3. To distinguish this equation from the tensor formulation of PPC.
4. To examine its dimensional consistency.
5. To describe its proposed weak-field application.
6. To explain the proposed use of four-acceleration in the strong-field regime through a separate Energy Conservation Law.
7. To identify the further mathematical derivations and physical tests needed to evaluate the proposal.

3. Fundamental PPC Relations

3.1 Mass density

Let ρ denote the mass density of matter.

In a spacetime-dependent formulation, mass density may vary with position and time:

$\rho = \rho(x, y, z, ct)$

The coordinate ct represents the time coordinate expressed in units of length, where c is the speed of light.

3.2 Energy density

The PPC framework adopts the mass-energy relation:

$E_d = \rho c^2$

Here, Ed denotes energy density, measured in joules per cubic metre, and ρ denotes mass density, measured in kilograms per cubic metre.

This relation is used as part of the proposed PPC framework.

3.3 Gravitational pressure

The existing PPC pressure relation is:

$P_g = \omega E_d$

Using $E_d = \rho c^2$:

$P_g = \omega \rho c^2$

The parameter ω may vary with spacetime. It should not be assumed to be a universal constant unless a particular PPC model or solution establishes that condition.

Consequently, gravitational pressure may also vary with spacetime:

$P_g = P_g(x, y, z, ct)$

The reference pressure, mass density, acceleration, characteristic curvature length, energy density, and proportionality parameter may likewise be spacetime-dependent quantities.

4. The Additional Scalar Equation

The additional equation proposed in this paper is:

$P_g = P_0 + \rho a L$

The symbols are defined as follows:

P_g = gravitational pressure

P_0 = reference pressure

ρ = mass density

a = magnitude of the relevant acceleration

L = characteristic length of curvature

The equation proposes that gravitational pressure consists of a reference-pressure contribution and an additional contribution proportional to mass density, acceleration magnitude, and characteristic curvature length.

The equation has two principal cases.

4.1 Case A: Zero reference pressure

If $P_0 = 0$, then:

$Pg = \rho a L$

This is the zero-reference-pressure case of the proposed scalar equation.

4.2 Case B: Non-zero reference pressure

If P_0 is non-zero, then:

$$P_g = P_0 + \rho a L$$

In this case, the total proposed gravitational pressure includes both the reference pressure and the acceleration–density–length contribution.

These cases are alternative conditions of the same proposed equation; they are not separate tensor equations.

5. Definition of the Characteristic Curvature Length

The symbol L denotes the characteristic length of curvature in the PPC model.

It represents a length scale selected to describe the curvature-related region or structure under consideration. It is not assumed to be a universal constant.

For a spacetime-dependent model:

$$L = L(x, y, z, ct)$$

The value of L may depend on the physical system, the curvature configuration, and the model's specified reference conditions.

6. Dimensional Analysis

The proposed scalar equation must be dimensionally consistent.

The relevant dimensions are:

Mass density, ρ: kg/m^3

Acceleration magnitude, a: m/s^2

Characteristic length, L: m

Reference pressure, P_0: Pa

Gravitational pressure, P_g: Pa

The dimensions of the product $\rho a L$ are:

$(kg/m^3) \times (m/s^2) \times (m)$

$= kg/(m \cdot s^2)$

Since:

$1\ Pa = 1\ kg/(m \cdot s^2)$

the product ρaL has the dimensions of pressure.

Therefore:

$P_g = P_0 + \rho aL$

is dimensionally consistent, provided that P_0 is also a pressure.

Dimensional consistency is a necessary condition for a physical equation, but it does not by itself prove that the equation correctly describes gravitational phenomena.

7. Relationship to the Existing PPC Pressure Relation

The existing PPC relation is:

$P_g = \omega \rho c^2$

The additional scalar equation is:

$P_g = P_0 + \rho aL$

If both relations are assumed to apply to the same physical state, they can be equated:

$\omega \rho c^2 = P_0 + \rho aL$

For $\rho \neq 0$, division by ρc^2 gives:

$\omega = P_0/(\rho c^2) + aL/c^2$

This expression describes the relationship between the two proposed pressure expressions under the assumption that both hold simultaneously.

For the special case $P_0 = 0$:

$\omega = aL/c^2$

If ω is constant in a particular solution, then:

$aL = \omega c^2$

This condition does not require a and L to be individually constant. They may vary while their product remains constant.

However, if the two pressure relations are intended for different regimes or different formulations, their equivalence should not be imposed without an explicit model connecting those regimes.

The relationship above is an algebraic consequence of assuming both equations apply to the same state. It is not an independent derivation of either equation.

8. Distinction from the Tensor Formulation

A central clarification of this paper is that the additional equation

$P_g = P_0 + \rho a L$

is proposed as a distinct scalar relationship within the broader PPC research programme.

It is not presented as a tensor equation. It is also not claimed to have been derived from the stress-energy tensor formulation or from the covariant conservation equation:

$\nabla_\mu T^{\mu\nu} = 0$

The tensor formulation and the additional scalar equation have different mathematical forms and roles in the proposed framework.

The tensor formulation uses tensor quantities and covariant relationships. The additional equation uses scalar quantities: pressure, reference pressure, mass density, acceleration magnitude, and characteristic length.

The use of a quantity obtained from four-acceleration in a scalar equation does not, by itself, turn that equation into a tensor equation.

Likewise, describing the scalar equation as distinct does not establish that it is automatically compatible with the tensor formulation. Compatibility, equivalence, or independence must be demonstrated through a separate mathematical analysis.

Accordingly, this paper treats the additional scalar equation as a separate PPC proposal whose relationship to the tensor formulation remains a subject for investigation.

9. Weak-Field Application

In a Newtonian weak-field description, gravitational acceleration can be written as:

$g = -\nabla \Phi$

where Φ is the Newtonian gravitational potential and g is the gravitational acceleration vector.

If the acceleration magnitude in the additional PPC equation is defined as:

a = |g|

then the scalar equation becomes:

P_g = P_0 + ρ|g|L

Equivalently:

P_g = P_0 + ρ|∇Φ|L

These expressions represent the proposed weak-field form under the stated definition of acceleration.

The vector g and its magnitude |g| should not be confused. The scalar equation uses an acceleration magnitude, while the Newtonian gravitational field is a vector.

The term ρg has the dimensions of force per unit volume. Multiplying it by a length produces a pressure scale. This dimensional relationship helps explain the structure of the proposed equation, but it does not establish that the resulting pressure field reproduces Newtonian gravity or the predictions of General Relativity.

To establish such a result, a specific gravitational potential, density distribution, reference pressure, and rule for selecting L would be needed. The equation would then need to be tested against known weak-field solutions and observations.

10. Strong-Field Application and Four-Acceleration

The author proposes a strong-field application involving four-acceleration through a separate Energy Conservation Law formulation.

Four-acceleration is conventionally represented by:

a^μ = u^v ∇v u^μ

where:

u^μ is the four-velocity,

∇_v denotes the covariant derivative, and

a^μ is the four-acceleration vector.

For a scalar pressure equation, a scalar magnitude may be required. With metric signature (−,+,+,+), one possible definition is:

$\alpha = \sqrt{(a^{\mu} a_{\mu})}$

This magnitude is appropriate when the four-acceleration is spacelike or zero, as it is for the four-acceleration of a physical timelike trajectory under the usual relativistic conventions.

If the author's Energy Conservation Law identifies this magnitude as the acceleration quantity to be used in the strong-field scalar equation, a possible form is:

$P_g = P_0 + \rho \alpha L$

Here, α denotes the magnitude of four-acceleration.

This expression is a proposed implementation, conditional on the definitions and derivation in the separate Energy Conservation Law. It should not be treated as a completed derivation until that law and its connection to the scalar equation are explicitly provided.

10.1 Strong-Field Application: Four-Acceleration Through the PPC Energy-Conservation Framework

In the strong-field regime, Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity proposes the use of a modified energy-pressure tensor within the energy-momentum conservation framework of General Relativity to formulate relativistic four-acceleration.

The PPC energy-pressure tensor is written as:

$T^{\mu\nu} = (E_d + P_g) u^{\mu} u^{\nu} + P_g g^{\mu\nu}$

where:

- $T^{\mu\nu}$ is the PPC energy-pressure tensor.
- E_d is the energy density.
- P_g is the gravitational pressure.
- u^{μ} is the four-velocity.
- $g^{\mu\nu}$ is the spacetime metric tensor.

The relativistic energy-momentum conservation equation is:

$\nabla_{\nu} T^{\mu\nu} = 0$

Applying this conservation equation to the stated PPC tensor gives the relativistic fluid equation:

$(E_d + P_g) a^{\mu} = -(g^{\mu\nu} + u^{\mu} u^{\nu}) \nabla_{\nu} P_g$

The four-acceleration is defined as:

$a^\mu = u^\nu \nabla_\nu u^\mu$

In this formulation, the pressure-gradient term is projected onto the directions orthogonal to the fluid four-velocity. When $E_d + P_g$ is non-zero, the equation can be rearranged as:

$a^\mu = -[(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g]/(E_d + P_g)$

Within the PPC interpretation, this relationship describes how the gravitational-pressure gradient contributes to the four-acceleration of a relativistic fluid.

The proposal uses the GR energy-momentum conservation framework together with the PPC energy-pressure tensor. It should therefore not be described as an acceleration law derived from an unrelated conservation principle.

The equation's physical implications depend on the precise definition of the PPC tensor, the metric and four-velocity conventions, and the assumptions concerning the fluid. In particular, the tensor's constitutive relations and any differences from the standard perfect-fluid formulation must be specified to establish which consequences are specific to PPC.

10.2 Relationship Between Four-Acceleration and the Additional Scalar Equation

The strong-field tensor formulation and the additional scalar equation are distinct mathematical relationships within the PPC framework, although they may be connected through the acceleration quantity.

The additional scalar equation is:

$P_g = P_0 + \rho a L$

where:

P_g = gravitational pressure;

P_0 = reference pressure;

ρ = mass density;

a = the specified scalar acceleration magnitude; and

L = characteristic length of curvature.

The tensor formulation expresses four-acceleration through the PPC energy-pressure tensor and the GR energy-momentum conservation equation. The additional scalar equation, by

contrast, proposes a relationship among gravitational pressure, reference pressure, mass density, acceleration magnitude, and characteristic curvature length.

The scalar equation is not itself a tensor equation, nor is it claimed to be identical to the tensor conservation equation.

However, the acceleration used in the scalar equation may be defined by reference to the four-acceleration obtained from the tensor formulation. If this is the intended connection, the model must specify how the four-vector a^μ is converted into the scalar quantity a.

For a timelike fluid trajectory, one possible definition under metric signature $(-,+,+,+)$ is the proper four-acceleration magnitude:

$$a = \sqrt{(a^\mu a_\mu)}$$

This definition is applicable when the four-acceleration is spacelike or zero. It should be distinguished from the magnitude of Newtonian gravitational acceleration and from coordinate acceleration.

With that definition, a possible strong-field form of the additional scalar equation is:

$$P_g = P_0 + \rho L\sqrt{(a^\mu a_\mu)}$$

This expression is conditional on the author's intended definition of a and on the physical meaning assigned to ρ and L in the strong-field model.

The two formulations should therefore be treated as distinct but potentially related components of PPC. Their compatibility must be demonstrated through explicit definitions, derivations, and consistency conditions rather than assumed merely because both contain gravitational pressure or acceleration.

11. Spacetime-Dependent Formulation

The additional scalar equation may be expressed as a relationship among fields defined throughout spacetime:

$$P_g(x, y, z, ct) = P_0(x, y, z, ct) + \rho(x, y, z, ct)a(x, y, z, ct)L(x, y, z, ct)$$

Here, x, y, and z represent spatial coordinates, while ct represents the time coordinate expressed in units of length.

The equation allows the quantities P_g, P_0, ρ, a, and L to vary with spacetime. Their variations must be determined by the physical assumptions and equations of the specific PPC model.

For a strong-field application in which the scalar acceleration is defined as the proper four-acceleration magnitude, the proposed relationship becomes:

$$P_g(x, y, z, ct) = P_0(x, y, z, ct) + \rho(x, y, z, ct)L(x, y, z, ct)\sqrt{(a^\mu a_\mu)}$$

where:

$$a^\mu = u^\nu \nabla_\nu u^\mu$$

The four-acceleration is obtained from the relativistic motion described by the PPC tensor/conservation formulation. The scalar pressure equation then uses its specified magnitude as an input.

This does not make the scalar equation a tensor equation. It establishes a possible connection between a scalar pressure relationship and a quantity calculated from the tensor formulation.

The complete model must define the coordinate and metric conventions, the matter or fluid trajectory, the reference pressure, the density, and the method used to determine L. It must also specify the conditions under which the scalar relationship is intended to apply.

The coordinate notation alone does not provide evolution equations for these fields. Those equations, together with appropriate initial or boundary conditions, are needed to calculate a unique physical solution.

12. Reference Pressure and Boundary Conditions

The reference pressure P_0 is an independent term in the proposed scalar equation:

$$P_g = P_0 + \rho aL$$

Its physical meaning must be defined within the PPC model. Possible interpretations include a selected reference value, a boundary pressure, or a background gravitational-pressure field. These interpretations should not be treated as equivalent unless the model establishes their relationship.

If P_0 is a spacetime-dependent field, its variation must be specified:

$$P_0 = P_0(x, y, z, ct)$$

The zero-reference-pressure case is:

$$P_0 = 0$$

Under this condition, the scalar equation reduces to:

$$P_g = \rho aL$$

This is a particular model condition. It does not, by itself, establish that gravitational pressure vanishes in vacuum or at infinity.

The reference pressure and boundary conditions must be selected consistently with the intended physical system. If the scalar equation is applied alongside the PPC tensor/conservation formulation, the model should explain how the reference pressure is defined and whether it is independently specified or constrained by other PPC equations.

The characteristic curvature length L must also be defined with sufficient precision to permit reproducible calculations. If L is intended to represent both a characteristic curvature length and the height of a gravitational well, the model must specify the geometric construction that connects those interpretations.

Finally, the pressure and acceleration used in the scalar equation must be distinguished from quantities defined by the tensor formulation. Any connection between them should be stated explicitly rather than inferred from their shared notation.

13. Mathematical and Physical Questions for Further Research

The additional scalar equation and the PPC tensor/conservation formulation require further mathematical specification and physical testing. The following questions identify the principal areas for investigation.

13.1 Definition of the PPC Tensor

The PPC energy-pressure tensor must be defined together with its metric signature, four-velocity normalization, and constitutive relations.

The author should identify which components or relations are modified relative to the standard perfect-fluid tensor and explain the mathematical and physical motivation for those modifications.

13.2 Derivation of Four-Acceleration

The derivation should explicitly substitute the proposed PPC tensor into:

$\nabla_v T^{\mu v} = 0$

The energy and momentum projections should then be presented with the conventions and assumptions used to obtain the four-acceleration equation.

The analysis should establish which results follow from the conservation equation and which depend on additional PPC assumptions. The case $E_d + P_g = 0$ also requires separate treatment because the rearranged expression for a^μ contains this quantity in the denominator.

13.3 Connection Between Tensor-Derived Acceleration and the Scalar Equation

The model should define how the acceleration used in:

$P_g = P_0 + \rho a L$

is obtained from the tensor/conservation formulation.

For a weak-field application, the acceleration may be defined as the magnitude of Newtonian gravitational acceleration. For a strong-field application, the author may use a properly defined magnitude of four-acceleration.

The two definitions are not interchangeable without a stated physical interpretation and a derivation of the relevant limiting relationship.

13.4 Definition of Reference Pressure

The model should establish whether P0 is fixed, prescribed at a boundary, determined by a background solution, or governed by an additional field equation.

The conditions under which $P_0 = 0$ is appropriate should be specified. A zero reference value should not automatically be interpreted as zero total gravitational pressure.

13.5 Consistency Between the Scalar and Tensor Formulations

The additional scalar equation and the tensor/conservation formulation should be assessed separately before their relationship is established.

The analysis should distinguish:

- The four-acceleration equation obtained from the PPC tensor and energy-momentum conservation framework;
- The additional assumption expressed by $P_g = P_0 + \rho a L$;
- The conditions under which both equations apply to the same physical state; and
- The constraints required for them to be mutually consistent.

If the additional scalar equation is used together with the existing PPC relation $P_g = \omega E_d$ and $E_d = \rho c^2$, their algebraic relationship can be derived under the assumption that all three relations hold simultaneously. That algebraic relationship does not by itself prove the physical validity of the additional equation.

13.6 Dimensional and Mathematical Consistency

The scalar equation is dimensionally consistent when ρ is mass density, a is acceleration magnitude, L is length, and P_0 and P_g are pressures.

Further checks must examine the definitions of the fields, their spacetime dependence, the tensor properties, and the assumptions used in the conservation-law derivation.

Dimensional consistency is necessary but is not sufficient to establish a physical law.

13.7 Weak-Field and Strong-Field Limits

The model should explain how its equations behave in weak gravitational fields and in the strong-field regime.

The weak-field analysis should specify the relationship between the scalar acceleration and Newtonian gravitational acceleration. The strong-field analysis should use the four-acceleration defined by the tensor/conservation formulation and state the relevant fluid and geometric assumptions.

The model should also examine cases with vanishing pressure gradients and cases in which the four-acceleration vanishes, without automatically equating zero four-acceleration with zero spacetime curvature.

13.8 Comparison with Established Relativistic Physics

The proposed formulation should be compared with standard relativistic perfect-fluid dynamics and General Relativity.

The paper should identify which mathematical expressions coincide with established equations, which interpretations are specific to PPC, and which results constitute genuinely new predictions.

Any claim of a new strong-field effect should follow from an explicit derivation rather than from the presence of four-acceleration alone.

13.9 Empirical Testing

Potential tests should be selected according to quantitative predictions derived from the PPC equations. Possible areas include relativistic fluid motion, compact astrophysical objects, orbital dynamics, and strong-field gravitational systems.

A meaningful test requires a defined physical model, calculated predictions, parameter values, uncertainty estimates where appropriate, and comparison with relevant observations or established theoretical results.

13.10 Status and Scope of the Additional Equation

The equation:

$P_g = P_0 + \rho a L$

is presented as an additional scalar proposal within the PPC framework.

The strong-field four-acceleration formulation is developed through the PPC energy-pressure tensor and the GR energy-momentum conservation framework. The scalar equation may use an appropriately defined acceleration from that formulation, but it remains a separate relationship.

The next stage of research is to specify the mathematical connection between these formulations, derive their consequences under explicit assumptions, and determine whether the resulting predictions are consistent with established physics and supported by empirical evidence.

14. Spacetime-Dependent Formulation

The additional equation can be written as a relationship among spacetime-dependent fields:

$P_g(x, y, z, ct) = P_0(x, y, z, ct) + \rho(x, y, z, ct)a(x, y, z, ct)L(x, y, z, ct)$

This notation makes explicit that the variables may change with spatial position and time.

For a proposed strong-field application using four-acceleration magnitude α, the expression becomes:

$P_g(x, y, z, ct) = P_0(x, y, z, ct) + \rho(x, y, z, ct)\alpha(x, y, z, ct)L(x, y, z, ct)$

These equations express the proposed scalar relationship point by point in spacetime, subject to the definitions of the relevant fields.

A complete field model would additionally need to specify the equations governing these fields, their boundary or initial conditions, and the method for calculating L.

15. Limitations

This paper presents the additional scalar equation as a theoretical proposal within the PPC framework. Dimensional consistency and algebraic relationships have been examined, but these do not establish the equation as a verified law of nature.

The paper does not claim that the additional scalar equation follows from the tensor formulation.

Further mathematical development, clear definitions of the reference pressure and characteristic length, and comparison with established theoretical and observational results are required.

16. Conclusion

This paper presents the additional scalar equation proposed within Pawan Upadhyay's Pressure–Curvature Law of Gravity:

$P_g = P_0 + \rho a L$

The equation expresses gravitational pressure as a reference-pressure contribution plus a term involving mass density, acceleration magnitude, and characteristic curvature length.

When $P_0 = 0$, the equation reduces to:

$P_g = \rho a L$

When P_0 is non-zero, the reference-pressure contribution remains:

$P_g = P_0 + \rho a L$

The equation is dimensionally consistent when its quantities are defined as specified. If it is assumed to hold alongside the existing PPC relation $Pg = \omega \rho c^2$ for the same physical state, the resulting algebraic relationship is:

$\omega = P_0/(\rho c^2) + aL/c^2$

The paper emphasizes that this additional scalar equation is distinct from the tensor formulation. It is not claimed to have been derived from stress-energy conservation.

For the strong-field regime, the author proposes using four-acceleration through a Energy Conservation Law. A scalar magnitude of four-acceleration may be used in the pressure equation if that choice is justified by the separate derivation and by the model's physical definitions.

The additional scalar equation, the tensor formulation, and the Energy Conservation Law should therefore be treated as distinct components of the PPC research programme until their precise mathematical relationships are established.

The next step is to provide complete derivations, define the characteristic curvature length and reference pressure operationally, and test the resulting predictions against established gravitational theory and observations.

Author's Statement

The equation $P_g = P_0 + \rho a L$ is presented as an additional scalar proposal within Pawan Upadhyay's Pressure–Curvature Law of Gravity. Its distinction from the tensor formulation is intentional.

The proposal is open to mathematical scrutiny, independent verification, and empirical testing.

Technical Point: Relationship Between the PPC Tensor Formulation and the Additional Scalar Pressure Equation

A central technical distinction in Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity is the distinction between the tensor-based energy-conservation formulation and the additional scalar gravitational-pressure equation.

The PPC energy-pressure tensor is proposed as:

$$T^{\mu\nu} = (E_d + P_g)u^{\mu}u^{\nu} + P_g g^{\mu\nu}$$

The relativistic energy-momentum conservation law is:

$$\nabla_{\nu} T^{\mu\nu} = 0$$

Applying the conservation law to the stated perfect-fluid tensor gives the relativistic fluid equation:

$$(E_d + P_g)a^{\mu} = -(g^{\mu\nu} + u^{\mu}u^{\nu})\nabla_{\nu} P_g$$

where:

$$a^{\mu} = u^{\nu}\nabla_{\nu} u^{\mu}$$

This formulation provides the proposed PPC framework for describing four-acceleration through the gravitational-pressure gradient in relativistic fluid dynamics.

Separately, the PPC framework proposes the additional scalar equation:

$$P_g = P_0 + \rho a L$$

where Pg is gravitational pressure, P0 is reference pressure, ρ is mass density, a is the specified scalar acceleration magnitude, and L is the characteristic length of curvature.

The scalar equation is not itself a tensor equation. Nevertheless, the acceleration quantity used in it may be connected to the four-acceleration obtained from the PPC tensor/conservation formulation, provided that the conversion from the four-vector to the scalar quantity is explicitly defined.

If a is defined as the proper four-acceleration magnitude, one possible definition, using metric signature (−,+,+,+), is:

$a = \sqrt{(a^\mu a_\mu)}$

The resulting scalar equation would then take the conditional form:

$P_g = P_0 + \rho L \sqrt{(a^\mu a_\mu)}$

This expression is a proposed connection between the two formulations, not an automatic consequence of the tensor conservation equation. Its physical validity depends on the definitions of the quantities, the assumptions of the model, and a demonstration that the scalar equation is consistent with the tensor-based dynamics.

The principal technical requirement is therefore to distinguish the derivation of four-acceleration from the PPC tensor and conservation law from the separate postulate relating gravitational pressure to reference pressure, mass density, acceleration magnitude, and characteristic curvature length. The mathematical relationship between these formulations should be established explicitly, and their predictions should be tested against established relativistic results and observations.

Clarification: Relationship Between the PPC Tensor Formulation and the Additional Scalar Equation

Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity proposes two distinct but potentially related mathematical formulations: a tensor-based energy-conservation formulation and an additional scalar gravitational-pressure equation.

The tensor formulation uses the PPC energy-pressure tensor:

$T^{\mu\nu} = (E_d + P_g)u^\mu u^\nu + P_g g^{\mu\nu}$

together with the relativistic energy-momentum conservation law:

$\nabla_\nu T^{\mu\nu} = 0$

Applying this conservation law to the stated perfect-fluid tensor gives the relativistic fluid equation:

$(E_d + P_g)a^\mu = -(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g$

where the four-acceleration is defined as:

$a^\mu = u^\nu \nabla_\nu u^\mu$

Within the PPC framework, this formulation describes how the gravitational-pressure gradient contributes to four-acceleration in relativistic fluid dynamics. The proposed strong-field

application therefore uses the energy-conservation framework of General Relativity with the PPC energy-pressure tensor.

Separately, the PPC theory proposes the additional scalar equation:

$P_g = P_0 + \rho a L$

where P_g is gravitational pressure, P_0 is reference pressure, ρ is mass density, a is the specified scalar acceleration magnitude, and L is the characteristic length of curvature.

This additional equation is distinct from the tensor formulation. It is not itself a tensor equation, nor is it claimed to be directly derived from the tensor conservation equation. However, the acceleration used in the scalar equation may be related to the four-acceleration obtained through the PPC tensor formulation, provided that the relationship is explicitly defined.

If a is intended to represent the proper four-acceleration magnitude, one possible definition, using metric signature $(-,+,+,+)$, is:

$a = \sqrt{(a^\mu a_\mu)}$

Under that specific definition, the scalar equation may be written as:

$P_g = P_0 + \rho L \sqrt{(a^\mu a_\mu)}$

This expression represents a possible connection between the two formulations, subject to the author's intended definitions and assumptions. It is not automatically implied by the conservation law.

The central clarification is that the PPC tensor formulation provides the proposed strong-field four-acceleration framework, while $P_g = P_0 + \rho a L$ is an additional scalar relationship. These formulations may be investigated together within the broader PPC theory, but their mathematical compatibility, physical interpretation, and empirical consequences must be established through explicit derivation and testing.

Separate Research Note: Gravitational Fluid Pressure in the PPC Law of Gravity

Author: Pawan Upadhyay
Research Framework: Pressure–Curvature (PPC) Law of Gravity

1. Purpose of This Note

This note introduces the concept of **gravitational fluid pressure** as a subject for investigation within Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity.

The concept explores a possible mathematical analogy between ordinary fluid pressure and the gravitational pressure proposed within PPC. It distinguishes the established mechanics of material fluids from the proposed interpretation of pressure associated with spacetime curvature.

2. Proposed Scalar Relationship

The additional scalar equation proposed within the PPC framework is:

$P_g = P_0 + \rho a L$

where:

- P_g represents gravitational pressure.
- P_0 represents reference pressure.
- ρ represents mass density.
- a represents the specified acceleration magnitude.
- L represents the characteristic length of curvature.

When the reference pressure is zero:

$P_g = \rho a L$

When the reference pressure is non-zero:

$P_g = P_0 + \rho a L$

The product $\rho a L$ has the dimensions of pressure when the quantities have their stated physical dimensions. This establishes dimensional consistency, but further derivation is needed to establish the physical validity of the proposed relationship.

3. Comparison with Fluid Pressure

For a static fluid of constant density under uniform gravitational acceleration, the hydrostatic equation is:

$P = P_0 + \rho g h$

Here, P is fluid pressure, P_0 is pressure at a reference level, ρ is fluid density, g is gravitational acceleration, and h is depth below the reference level.

The hydrostatic equation and the additional PPC equation have similar mathematical structures. In particular, both contain a reference-pressure term and a density–acceleration–length contribution.

However, h represents fluid depth, whereas L represents the characteristic length of curvature in the proposed PPC model. Their mathematical resemblance does not establish that they are physically identical.

4. Variable Quantities in Proposed Gravitational Fluid Pressure

In the proposed gravitational fluid pressure formulation of Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity, all physical quantities are treated as variable fields in spacetime, except the speed of light in vacuum, c, which is taken as constant.

The additional scalar equation is:

$P_g = P_0 + \rho a L$

In this equation, the quantities are defined as follows:

- P_g = variable gravitational pressure.
- P_0 = variable reference pressure.
- ρ = variable mass density.
- a = variable acceleration magnitude.
- L = variable characteristic length of curvature.
- c = constant speed of light in vacuum.

The spacetime dependence of the variable quantities can be expressed as:

$P_g = P_g(x, y, z, ct)$

$P_0 = P_0(x, y, z, ct)$

$\rho = \rho(x, y, z, ct)$

$a = a(x, y, z, ct)$

$L = L(x, y, z, ct)$

The speed of light in vacuum, c, is treated as a constant parameter rather than a variable field in this formulation.

The equation therefore represents a local relationship among quantities that may change with spatial position and time. Their variations must be specified by the physical assumptions and mathematical equations of the particular PPC model.

The statement that all quantities except c are variable expresses the proposed framework's modelling assumption. It does not require every quantity to change at every point or at every moment; particular solutions may contain constant quantities under specified conditions.

5. Relationship to the PPC Tensor Formulation

The PPC framework proposes an energy-pressure tensor:

$T^{\mu\nu} = (E_d + P_g)u^{\mu}u^{\nu} + P_g g^{\mu\nu}$

together with the energy-momentum conservation equation:

$\nabla_{\nu} T^{\mu\nu} = 0$

For the stated perfect-fluid tensor, the conservation equation gives a relativistic fluid equation involving four-acceleration and the pressure gradient.

The additional scalar relationship, $P_g = P_0 + \rho aL$, is distinct from this tensor formulation. The acceleration used in the scalar equation may be connected to the four-acceleration obtained through the tensor/conservation formulation, provided that the connection is explicitly defined.

The scalar equation is therefore an additional proposed relationship, not a substitute for the tensor conservation equation.

6. Questions for Further Investigation

Further research should address the following questions:

1. What physical quantity does gravitational fluid pressure represent?
2. How is the characteristic curvature length L determined?
3. How does the proposed scalar equation relate to the PPC tensor formulation?
4. How is the acceleration magnitude defined in weak-field and strong-field applications?
5. Can the resulting model reproduce established gravitational results and yield independently testable predictions?

7. Conclusion

Gravitational fluid pressure is a proposed concept for investigating the relationship between fluid-pressure mathematics and the pressure–curvature interpretation of gravitation in PPC.

The hydrostatic equation provides a useful mathematical comparison, while the additional scalar equation introduces a distinct relationship involving gravitational pressure, reference pressure, mass density, acceleration magnitude, and characteristic curvature length.

The next step is to define these quantities operationally, establish the relationship between the scalar and tensor formulations, and determine through mathematical derivation and empirical testing whether the proposed concept provides a physically meaningful description of gravitation.

Status: This note presents a theoretical proposal for further investigation, not an experimentally established law of physics.

Separate Research Note: Gravitational Fluid Pressure in the Pressure–Curvature (PPC) Law of Gravity

1. Purpose of This Research Note

This research note presents the proposed concept of gravitational fluid pressure within Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity. It introduces a proposed scalar pressure relationship, describes the variability of its quantities across spacetime, and discusses its potential relationship to the PPC energy-density relations and stress-energy tensor.

An important distinction underlies this proposal: **spacetime is not an ordinary fluid**. The expression *gravitational fluid pressure* denotes a proposed theoretical concept within the PPC framework; it does not mean that spacetime is composed of an ordinary material fluid.

The equations in this note are presented as proposed relationships whose mathematical consistency and physical validity require further investigation.

2. Proposed Scalar Relationship for Gravitational Fluid Pressure

The proposed scalar equation is:

$P_g = P_0 + \rho a L$

where:

- **P_g** = proposed gravitational fluid pressure.
- **P_0** = reference gravitational pressure.
- **ρ** = mass density.
- **a** = specified acceleration magnitude.
- **L** = characteristic length of curvature.

The proposed relationship expresses gravitational pressure as the sum of a reference pressure and a term involving mass density, acceleration, and characteristic curvature length.

The physical meaning of each quantity must be defined precisely. In particular, the model must specify how gravitational pressure is interpreted, how the characteristic length L is determined, and which definition of acceleration is used.

2.1 Special Case: Zero Reference Pressure

If the reference pressure is zero,

$P_0 = 0$

the proposed equation reduces to:

$P_g = \rho a L$

Under this condition, the proposed gravitational pressure depends on the mass density, acceleration magnitude, and characteristic length.

2.2 General Case: Nonzero Reference Pressure

If the reference pressure is nonzero,

$P_0 \neq 0$

the full relationship remains:

$P_g = P_0 + \rho a L$

The reference pressure may itself be a variable field within the general PPC formulation. Its physical interpretation and value in a particular system must be established through the model's assumptions and boundary conditions.

3. Dimensional Consistency of the Proposed Equation

The SI units of the relevant quantities are:

- Mass density, ρ: kg/m^3
- Acceleration, a: m/s^2

- Characteristic length, L: m
- Pressure, P_g and P_0: pascals (Pa)

The dimensions of the product ρaL are:

$(kg/m^3)(m/s^2)(m) = kg/(m \cdot s^2) = Pa$

Thus, the term ρaL has the dimensions of pressure, and the proposed equation is dimensionally consistent when its quantities have the definitions and units specified above.

Dimensional consistency is a necessary mathematical condition. It does not, by itself, establish that the equation describes an actual gravitational phenomenon.

4. Variable Quantities in Proposed Gravitational Fluid Pressure

Within the proposed PPC formulation, **all quantities in the gravitational fluid pressure relationship are treated as variable except the speed of light, c, which is taken as constant**.

The spacetime coordinates are represented as (x, y, z, ct), where c is the constant speed of light in vacuum and t represents time.

The proposed quantities may be expressed as spacetime-dependent fields:

P_g = P_g(x, y, z, ct)

P_0 = P_0(x, y, z, ct)

ρ = ρ(x, y, z, ct)

a = a(x, y, z, ct)

L = L(x, y, z, ct)

The complete scalar relationship can therefore be written as:

Pg(x, y, z, ct) = P0(x, y, z, ct) + ρ(x, y, z, ct)a(x, y, z, ct)L(x, y, z, ct)

In this formulation, c is treated as constant, while P_g, P_0, ρ, a, and L are permitted to vary with position and time.

This does not require every variable to change at every spacetime point. Particular solutions may contain constant fields or quantities that remain unchanged under specified physical conditions. The general proposal permits variability, while the governing equations and boundary conditions would determine the actual behavior in each solution.

The model must ultimately establish which fields are independent, how they are related, and what equations govern their spacetime dependence.

5. Distinction Between Gravitational Fluid Pressure and Ordinary Fluid Pressure

The PPC concept of gravitational fluid pressure must be distinguished from conventional pressure in an ordinary material fluid.

Water, air, and other conventional fluids are physical substances with material properties. The term *gravitational fluid pressure*, as used in this proposal, does not assert that spacetime is such a substance.

The proposed PPC equation is:

$$P_g = P_0 + \rho a L$$

For comparison, a conventional hydrostatic pressure relationship under appropriate assumptions is:

$$P = P_0 + \rho g h$$

where h is the fluid depth measured relative to a reference level and g is the magnitude of gravitational acceleration.

The two expressions share a similar mathematical structure. However, their mathematical resemblance does not establish that they represent the same physical mechanism.

In the hydrostatic equation, h represents a fluid depth. In the PPC proposal, L represents a characteristic length associated with curvature. These quantities have different intended interpretations and cannot automatically be treated as equivalent.

Accordingly, the term *gravitational fluid pressure* should be understood as terminology for a proposed theoretical quantity within the PPC framework. **It does not mean that spacetime is an ordinary fluid, nor does the hydrostatic analogy establish that spacetime behaves like one.**

6. Relationship to PPC Energy Density and Pressure

The PPC framework also proposes the following relationships:

$$E_d = \rho c^2$$

and

$P_g = \omega E_d$

where E_d denotes the proposed energy density and ω is a dimensionless equation of state parameter.

Combining these relationships gives:

$P_g = \omega \rho c^2$

If this expression and the proposed scalar equation

$P_g = P_0 + \rho a L$

are assumed to describe the same physical state, they must satisfy:

$\omega \rho c^2 = P_0 + \rho a L$

For $\rho \neq 0$, this can be rearranged as:

$\omega = P_0/(\rho c^2) + aL/c^2$

When $P_0 = 0$, the expression simplifies to:

$\omega = aL/c^2$

These equations establish a conditional algebraic relationship between the proposed scalar pressure equation and the proposed energy-density relationship.

If ω varies with spacetime, its variation must be compatible with the other fields. If ω is constant in a particular solution and $P_0 = 0$, the product aL must satisfy $aL = \omega c^2$. This condition does not require a and L to be individually constant.

These relationships describe mathematical compatibility under the stated assumptions. They do not independently establish the physical validity of the underlying PPC equations.

7. Relationship to the PPC Stress-Energy Tensor

The PPC framework has also been expressed through a proposed stress-energy tensor:

$T^{\mu\nu} = (E_d + P_g)u^\mu u^\nu + P_g g^{\mu\nu}$

with the conservation equation:

$\nabla_\nu T^{\mu\nu} = 0$

Here:

- T^μv is the stress-energy tensor in the stated formulation.
- E_d is the proposed energy density.
- P_g is the proposed pressure.
- u^μ is the four-velocity.
- g^μv is the inverse spacetime metric.
- ∇_v denotes the covariant derivative.

Under the usual perfect-fluid assumptions and compatible metric-signature and index conventions, projecting stress-energy conservation perpendicular to the four-velocity gives:

(E_d + P_g)a^μ = −(g^μv + u^μu^v)∇vPg

where the four-acceleration is defined as:

a^μ = u^v∇vu^μ

This equation describes the relativistic fluid-dynamical form associated with the stated tensor and conservation equation. It does not, by itself, establish the additional scalar relationship:

P_g = P_0 + ρaL

The scalar relationship should therefore be treated as a separate proposed relation unless a mathematical derivation explicitly establishes how it follows from, or consistently supplements, the tensor formulation.

7.1 Defining Acceleration in the Scalar Equation

The acceleration a in the scalar equation must be defined precisely.

If the intended quantity is the magnitude of the proper four-acceleration, then, using metric signature (−,+,+,+), one possible definition is:

a = √(a^μa_μ)

This represents the magnitude of four-acceleration for a timelike trajectory. It is not automatically identical to a locally measured three-acceleration or to the magnitude of a gravitational field.

Consequently, a PPC application must specify which acceleration enters the scalar pressure equation and explain how that quantity is obtained in the relevant physical setting.

The tensor formulation and the scalar relationship may be investigated together, but their compatibility requires explicit mathematical demonstration. The scalar equation should not be described as a consequence of stress-energy conservation unless such a derivation has actually been established.

8. Questions for Further Mathematical and Physical Investigation

The following issues remain important for developing the proposal:

1. **Reference pressure:** What determines P_0, and how is it calculated or measured?

2. **Mass density:** What physical system does ρ describe, and how is it related to the proposed gravitational pressure?

3. **Acceleration:** Is a a three-acceleration, a gravitational-field acceleration, a proper four-acceleration, or another defined quantity?

4. **Characteristic length:** What exact geometric or physical procedure determines L?

5. **Spacetime variability:** What equations govern the variation of P_g, P_0, ρ, a, and L?

6. **Tensor compatibility:** Under what conditions can the scalar pressure relationship coexist with the proposed PPC stress-energy tensor and conservation equation?

7. **Consistency with established results:** Does the formulation reproduce relevant tested predictions of established gravitational theories in their applicable regimes?

8. **Observable predictions:** Does the PPC proposal yield measurable predictions that distinguish it from existing theories?

9. **Boundary and initial conditions:** What conditions determine the fields in a specific physical system?

Answering these questions would help establish whether the proposed relationship can be developed into a mathematically consistent and empirically testable model.

10. Conclusion

The proposed scalar relationship for gravitational fluid pressure in the Pressure–Curvature (PPC) Law of Gravity is:

$P_g = P_0 + \rho a L$

The equation is dimensionally consistent when the quantities are assigned their stated SI units. In the proposed formulation, P_g, P_0, ρ, a, and L may vary with spacetime, while c, the speed of light in vacuum, is treated as constant.

The term *gravitational fluid pressure* is a theoretical designation within the PPC framework. It does not imply that spacetime is an ordinary material fluid. The resemblance between the proposed equation and the conventional hydrostatic pressure equation is a mathematical analogy, not proof of a shared physical mechanism.

The proposed scalar equation, the PPC energy-density relations, and the PPC stress-energy tensor must be connected through explicit definitions and mathematical derivations. Their physical significance must then be evaluated through comparison with established gravitational results and empirical observations.

The PPC gravitational fluid pressure relationship is therefore a proposed theoretical component whose further development requires precise definitions, mathematical consistency, and testable physical predictions.

Definition: Characteristic Length of Curvature in the PPC Law of Gravity

1. General Definition

The **Characteristic Length of Curvature (L)** is the total length of a specified curvature path associated with the curvature formation of a physical system. The path may be circular, noncircular, or another geometrically defined curve, depending on the configuration of the system.

The characteristic length is not restricted to a circular circumference. Its geometric form and value depend on the curvature path defined by the proposed PPC model.

2. Mathematical Representation

For a circular curvature path:

$L = 2\pi R_c$

For a noncircular curvature path, L is the total length measured along the specified curve. In a suitable geometric representation, it may be calculated by integrating the differential arc length along that curve.

Thus, the general definition is the total length of the selected curvature path, while $L = 2\pi R_c$ is a special case for a circle.

3. SI Unit

The SI unit of L is the **meter (m)**, whether the specified curvature path is circular or noncircular.

4. Role in the PPC Scalar Equation

The proposed gravitational-pressure equation remains:

$P_g = P_0 + \rho a L$

Here, P_g is gravitational pressure, P_0 is reference pressure, ρ is mass density, a is the specified acceleration magnitude, and L is the characteristic length of curvature.

Because L has the dimension of length, the term $\rho a L$ has the dimension of pressure. This establishes dimensional consistency, but does not by itself validate the equation as a physical law.

5. Geometric Specification

For consistent application, the PPC model must specify how the curvature path is identified, how its total length is calculated, and how that path relates to the physical system. Circular geometry should not be assumed when the curvature formation is noncircular.

The characteristic length may be treated as variable when the selected path changes with the physical configuration, position, or time, provided those dependencies are defined by the model.

6. Conclusion

In the proposed PPC framework, L denotes the total length of a specified curvature path, whether circular or noncircular. The circular expression $L = 2\pi R_c$ is one special case, not a universal definition. The general geometric definition and its physical application require further specification and testing.

Separate Research Note: Characteristic Length of Curvature in the PPC Law of Gravity for a Circular Curvature Path

1. Introduction

In the proposed Pressure–Curvature (PPC) Law of Gravity, the Characteristic Length of Curvature, denoted by L, represents a length associated with the curvature formation of a physical system. When the specified curvature path is circular, L is defined as the **total circumference of the circular curvature path**, rather than its radius, diameter, or a straight-line distance.

This note describes the circular case of the proposed characteristic-length concept and its role in the PPC gravitational-pressure equation.

2. Definition of Circular Characteristic Length

For a circular curvature path, the characteristic length of curvature is the complete circumference of the circle.

The mathematical relation is:

$L = 2\pi R_c$

Where:

- **L** = total circular length of curvature, measured in meters (m).
- **R_c** = radius of the specified circular curvature path, measured in meters (m).
- **π** = mathematical constant, approximately 3.14159.

The circumference includes the entire circular path. Therefore, L is not equal to the radius or diameter.

3. Incorporation into the PPC Equation

The proposed scalar gravitational-pressure equation is:

$P_g = P_0 + \rho a L$

For a circular curvature path, substituting $L = 2\pi R_c$ gives:

$P_g = P_0 + 2\pi \rho a R_c$

Where:

- **P_g** = proposed gravitational pressure.
- **P_0** = reference pressure.
- **ρ** = mass density.
- **a** = specified acceleration magnitude.

- **R_c** = radius of the specified circular curvature path.

This expression represents the proposed additional pressure contribution in terms of mass density, acceleration magnitude, and the total circular length associated with the selected curvature path.

4. SI Unit and Dimensional Consistency

The SI unit of the characteristic length L is the **meter (m)**.

The dimensions of the additional pressure term are:

ρ: kg/m^3

a: m/s^2

L: m

Therefore:

$\rho aL = kg/(m \cdot s^2) = Pa$

The result has the SI unit of pressure, the pascal (Pa). Accordingly, the proposed equation is dimensionally consistent when P_g and P_0 are both pressures.

Dimensional consistency alone, however, does not demonstrate that the proposed equation correctly describes physical gravitational phenomena.

5. Geometric Interpretation

The circular characteristic length is determined by the radius of the specified circle. If R_c changes, the circumference L changes proportionally.

For the PPC model to be applied consistently, the relevant circular curvature path and its radius must be defined using a clear geometric prescription. The choice of this path should not be arbitrary or determined solely by the desired value of gravitational pressure.

This definition applies specifically to a circular curvature path. If curvature forms along a noncircular path, its characteristic length must instead be calculated using the total length of that specified path.

6. Conclusion

For a circular curvature path in the proposed PPC Law of Gravity, the Characteristic Length of Curvature is defined as:

$L = 2\pi R_c$

The resulting gravitational-pressure expression is:

$P_g = P_0 + 2\pi \rho a R_c$

Here, L is the total circumference, measured in meters. This definition provides a precise mathematical expression for the circular case, while the physical identification of the relevant curvature path and the empirical validity of the PPC pressure equation remain subjects for further investigation.

Short Note: Characteristic Length of Curvature in the PPC Law of Gravity

In the proposed Pressure–Curvature (PPC) Law of Gravity, the **Characteristic Length of Curvature (L)** represents the total length of a specified curvature path. Its geometric form depends on how the curvature is formed in the physical system; therefore, the definition is not restricted to circular paths.

For a circular curvature path, L is the total circumference:

$L = 2\pi R_c$

For a noncircular curvature path, L represents the total length along the specified path, determined by its geometry.

The proposed gravitational-pressure equation is:

$P_g = P_0 + \rho a L$

Here, P_g is gravitational pressure, P_0 is reference pressure, ρ is mass density, and a is the specified acceleration magnitude. The SI unit of L is the meter (m).

Thus, the circumference-based expression is a **special case** of the general characteristic-length concept. The geometric rule for identifying the relevant curvature path and calculating its length must be specified for each physical application.

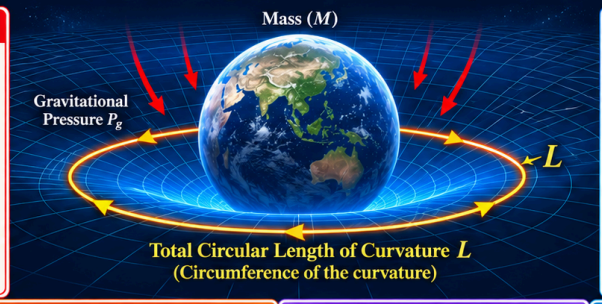
Pressure–Curvature (PPC) Law of Gravity

Gravitational Pressure, Total Circular Length of Curvature, and Strong-Field Four-Acceleration

Main Scalar Equation (PPC)

$$P_g = P_0 + \rho a L$$

- P_g = gravitational pressure
- P_0 = reference pressure
- ρ = mass density
- a = specified acceleration magnitude
- L = total circular length of curvature (variable, not constant)



Characteristic Length of Curvature (L)

- L is the total circular length of curvature (circumference) at a given radial distance.
- It is the complete circular length around the mass at that curvature.
- It is not a universal constant.
- It is distinct from any gravitational-well height.
- It must be defined for each physical system.
- It may vary with position and time.

$$L = 2\pi R_c$$

where R_c is the curvature radius at the considered circular path.

Weak-Field Application

Newtonian gravitational acceleration:

$$g = -\nabla\Phi$$

Magnitude:

$$a = |g|$$

Total circular length of curvature:

$$L = 2\pi R_c$$

PPC scalar equation (weak-field):

$$P_g = P_0 + \rho a L$$


Weak gravitational field (smaller curvature)

Strong-Field Application

Four-acceleration:

$$a^\mu = u^\nu \nabla_\nu u^\mu$$

Magnitude (proper four-acceleration):

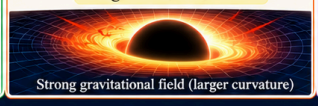
$$\alpha = \sqrt{a^\mu a_\mu}$$

Total circular length of curvature:

$$L = 2\pi R_c$$

PPC scalar equation (strong-field):

$$P_g = P_0 + \rho a L$$


Strong gravitational field (larger curvature)

PPC Energy-Density Relation

Mass–energy relation:

$$E_d = \rho c^2$$

Gravitational pressure relation:

$$P_g = \omega E_d$$

Therefore:

$$P_g = \omega \rho c^2$$

Compatibility with scalar equation:

$$\omega = \frac{P_0}{\rho c^2} + \frac{aL}{c^2}$$

The parameter ω may be constant or variable, depending on the model.

PPC Stress–Energy Tensor

Perfect-fluid form:

$$T^{\mu\nu} = (E_d + P_g)u^\mu u^\nu + P_g g^{\mu\nu}$$

Conservation equation:

$$\nabla_\nu T^{\mu\nu} = 0$$

Relativistic fluid equation:

$$(E_d + P_g)a^\mu = -(g^{\mu\nu} + u^\mu u^\nu)\nabla_\nu P_g$$

The scalar equation and the tensor equation are treated as mathematically distinct components of the PPC research programme.

All quantities (except c) may vary with spacetime: $P_g(x, y, z, ct)$, $P_0(x, y, z, ct)$, $\rho(x, y, z, ct)$, $a(x, y, z, ct)$, $L(x, y, z, ct)$